Lecture 1: Data Structure and Baseline Forecasts

ECON 417 Business Forecasting

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A Forecast Answers a Future Question

Definitions

A **forecast** is a statement about an outcome made before that outcome is observed.

A **model** is a rule connecting available information to an outcome.

Imagine: At 6 p.m. on Monday, you must plan Tuesday's sandwich supply.

- A **variable** is a quantity that can change: daily sales.
- An **observation** is one recorded value: Monday sold 120 sandwiches.
- The **target** is what you want to predict: Tuesday's sales.

Specify the Date, Horizon, and Unit

- **Forecast origin**: when you make the prediction.
- **Forecast horizon**: how far ahead you look.
- **Unit**: what the number measures.

Origin	Monday, 6 p.m.
Horizon	One day ahead: Tuesday
Target and unit	Tuesday sales, in sandwiches

Evaluation means comparing the saved prediction with the later outcome.

Actual, Fitted, and Forecast Values

Fitting means choosing a model's adjustable numbers using recorded data.

y_t	Actual recorded outcome in period t
$\hat{y}_t$	Fitted value for an observation in the data being fitted
$\hat{y}_{t+1\|t}$	Forecast made at t for the next period

Period t means the current day, month, or other chosen time unit.
In the forecast label, $t+1$ is the destination; t after the bar is the origin.

Close the Information Folder at the Origin

Information set

The facts available when you make the forecast, written as $\mathcal{I}_t$.

Available Monday evening

- Sales through Monday
- Tuesday promotion already announced
- Weather forecast issued Monday

Not available yet

- Tuesday's actual sales
- Tuesday's realized rainfall
- Wednesday's file correction

Leakage: Using Information Too Early

Data leakage

Information that should be unavailable enters the model's construction or evaluation.

Imagine: A Monday forecast uses Tuesday's actual sales.

- The recorded score can look excellent.
- The manager cannot use that information on Monday.

Key message: Check when each input became available.

Cross Section: Many Entities, One Date

Data are recorded observations. An **entity** is the shop, person, or other unit observed.

Cross-sectional data

Different entities observed for the same date or reference period.

Shop	Monday sales
A	120
B	90
C	150

We can compare shops on Monday. We cannot trace Shop A through time from this table alone.

Time Series: One Entity Across Dates

Time-series data

A quantity recorded in time order for one entity or a defined total.

	Saturday	Sunday	Monday
Shop A	100	110	130

- **Frequency**: how often observations are recorded, such as daily or monthly.
- One period ahead means one day for daily data, one month for monthly data.
- Reversing the order would turn this increase into a decrease.

Panel: Many Entities Across Dates

Panel data

Multiple identified entities followed through time.

Shop	Saturday	Sunday	Monday
A	100	110	130
B	80	85	90
C	140	135	150

Rows: follow one shop over time. **Columns**: compare shops on one date.

Start with a Simple Reference

Baseline forecast

A simple forecasting rule used as the reference for judging another model.

Observed sales: Saturday 100, Sunday 110, Monday 130.

- Use the latest value.
- Use an average of earlier values.

Key message: A more complicated model must earn its extra complexity.

All shop numbers in this lecture are illustrative.

Latest-Value Forecast

Naive forecast

Use the most recent observed value as the next prediction.

$$\hat{y}_{t+1|t} = y_t.$$

Monday sales were 130. Tuesday's forecast is **130 sandwiches**.

Persistence: high or low values tend to carry forward through time.

Historical-Mean Forecast

Mean

Add the values and divide by how many there are. A bar marks an average.

$$\hat{y}_{t+1|t} = \bar{y}_t, \qquad \bar{y}_t = \frac{y_1 + \cdots + y_t}{t}.$$

Use only observations available through the forecast origin.

$$\frac{100 + 110 + 130}{3} = 113.33 \text{ sandwiches.}$$

Moving Average: Keep a Fixed Window

Moving average and window length

Average a fixed number of the most recent observations. That number is the window length.

$$\hat{y}_{t+1|t} = \frac{y_t + y_{t-1} + y_{t-2}}{3}.$$

A three-day window uses 100, 110, 130; then 110, 130, 140.

Smoothing: reducing short-lived up-and-down movement by averaging.

Rolling and Expanding Averages Respond Differently

Expanding window: keep earlier observations and add each new one.

After 140 arrives	Values used	Average
Three-day moving window	110,130,140	126.67
Expanding window	100,110,130,140	120

- A longer history can reduce the influence of an unusual day.
- Old low values can slow the response to a genuinely higher level.

Forecast Error Has a Direction

Forecast error

Actual outcome minus the saved forecast.

$$e_{t+1} = y_{t+1} - \hat{y}_{t+1|t}.$$

Actual sales	Forecast	Error
125	130	-5
125	113.33	$+11.67$

Negative error: predicted too much. Positive error: predicted too little.

One Day Is Not Enough to Judge a Model

Save forecasts before the outcomes arrive, then collect their errors.

Day	Forecast	Actual	Error
1	100	110	10
2	120	115	-5
3	110	125	15

The average signed error is $(10 - 5 + 15)/3 = 6.67$.
Errors of opposite signs can cancel. We also need the typical **size** of a miss.

MAE: Average the Size of the Misses

Absolute value means distance from zero: $|-5| = 5$.
Mean absolute error (MAE) is the average absolute forecast error.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|e_i|.$$

N counts the evaluated forecasts; i labels one forecast; $\sum$ means add the terms.

$$MAE = \frac{10 + 5 + 15}{3} = 10 \text{ sandwiches.}$$

RMSE: Give Large Misses More Weight

Squaring means multiplying a number by itself.
Root mean squared error (RMSE): square the errors, average, then take the square root.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2}.$$

A square root reverses squaring and returns the result to the original unit.

$$\sqrt{\frac{10^2 + (-5)^2 + 15^2}{3}} = \sqrt{116.67} \approx 10.80 \text{ sandwiches.}$$

The Error Measure Reflects What You Care About

Miss size	Absolute contribution	Squared contribution
10	10	100
20	20	400

- Doubling the miss doubles its absolute contribution.
- Doubling the miss makes its squared contribution four times as large.
- MAE and RMSE do not separately price shortages and leftovers.

Key message: Choose a measure suited to the decision.

Training and Test Samples Have Different Jobs

A **sample** is a set of observations used for a calculation.

- **Training sample**: observations used to fit the model.
- **Test sample**: later observations reserved for checking forecasts.
- **In-sample performance**: performance on fitting data.
- **Out-of-sample performance**: performance on observations excluded from fitting.

Key message: A forecasting test must also respect what was known at each origin.

Make the Forecast Before Revealing the Answer

Imagine: You have January through June sales.

1. Fit the rule with January through April.
2. Save May's forecast before seeing May sales.
3. Observe May, record the error, then forecast June.

A **holdout** is a later period kept aside for evaluation.

Using its scores repeatedly to change the model turns it into **validation**: data used to help choose the model.

Compare the Same Forecasting Task

For a fair comparison, keep these the same:

- Target and measurement unit
- Horizon and forecast dates
- Information allowed at each origin

Two scores that cannot rank two models

10 sandwiches of daily sales error versus \$5,000 of monthly hospital revenue error.

Same-period co-movement can describe a relationship without being available for a forecast made earlier.

Compare Errors with a Feasible Baseline

A **benchmark** is the reference used for comparison. Here it is a baseline forecast that could actually have been made.

$$R^2_{\mathrm{OS}} = 1 - \frac{\sum_i (y_i - \hat{y}_i^{\mathrm{model}})^2}{\sum_i (y_i - \hat{y}_i^{\mathrm{baseline}})^2}.$$

OS means out of sample. Both totals use the same target and dates.

Model total 80, baseline total 100: score $1 - 80/100 = 0.20$.

Model total 120, baseline total 100: score -0.20, worse than the baseline.

The baseline error total must be positive. A full test-period mean was not known at its start.

Regression Learns a Relationship

Definitions

A **predictor** is an input supplied to help predict the target.

Regression describes a target using a systematic part related to predictors and an unexplained part.

A **coefficient** is a number governing that relationship; estimating it means choosing it from data.

Imagine: Planned lunch orders known today help predict tomorrow's sandwich sales.

Explain Each Part of the Regression

$$y_{t+1} = a + bx_t + u_{t+1}.$$

- y_{t+1}: tomorrow's actual sales.
- x_t: planned orders known today.
- **Intercept** a: systematic value when the input is zero.
- **Slope** b: change in the prediction per additional input unit.
- **Error term** u_{t+1}: the unexplained part of tomorrow's outcome.

Use the Known Input to Make a Forecast

Suppose the estimated intercept is 80 and the estimated slope is 2.
Today you know there are 20 planned orders.

$$\hat{y}_{t+1|t} = 80 + 2 \times 20 = 120.$$

- The prediction starts at 80 sandwiches.
- Each extra planned order adds 2 to the fitted prediction.
- Twenty planned orders give a forecast of 120 sandwiches.

An association alone does not prove that changing one input causes that outcome change.

Residual and Forecast Error: Same Subtraction, Different Timing

Residual

Observed outcome minus fitted value for an observation in the fitting data.

	Compare	When?
Residual	Actual with fitted value	On fitting observations
Forecast error	Actual with saved forecast	After the future outcome arrives

The model's underlying error term is unobserved. A calculated residual estimates an unexplained part.

A Lag Is an Earlier Value

Lag: an earlier value of a variable. Yesterday's sales are a one-period lag of today's sales.

Autoregression

Regression using the target's own past values as predictors. "Auto" refers to the series itself.

Pair	Earlier sales	Next sales
Saturday to Sunday	100	110
Sunday to Monday	110	130

AR(1) uses one lag. AR(2) uses two. AR(p) uses p lags.

An AR(1) Forecast Uses Today to Predict Tomorrow

$$y_{t+1} = c + \phi y_t + u_{t+1}.$$

- c: intercept.
- ϕ: persistence coefficient, describing how the current value carries forward.
- u_{t+1}: the unexplained future part.

With estimated $c = 20$ and $\phi = 0.8$, today's sales of 130 give

$$\hat{y}_{t+1|t} = 20 + 0.8 \times 130 = 124.$$

The future error has mean zero given the model's information; its realized value is unknown today.

A Forecast Informs a Decision

Uncertainty: more than one future outcome remains possible.
Probability: a number between zero and one describing the chance of an outcome.

- A **point forecast** is one predicted number.
- An **interval forecast** gives a range with a stated method or probability interpretation.

Imagine: A point forecast is 120; sales might be 100 or 140.
Ordering 120 can leave 20 portions or create a shortage of 20.

Key message: The costs of shortages and leftovers help determine the order.

Expected Value Averages Possible Outcomes

Expected value

An average of possible outcomes weighted by their probabilities: multiply each outcome by its chance and add.

Imagine: Sales are 100 or 140, each with probability one half.

$$0.5 \times 100 + 0.5 \times 140 = 120.$$

The average is 120 even though neither possible outcome equals 120.

Conditional Mean: Average Given What Is Known

Conditional expectation

The probability-weighted average of outcomes given specified information.

$$E[y_{t+1} \mid \mathcal{I}_t].$$

The outcome inside is tomorrow's sales. The information after the bar is what we know today.

Squared-error loss scores a miss by its square. Under this rule, the conditional mean minimizes expected squared error when the outcome distribution is known.

In practice, we estimate relationships from limited data.

Complete One Forecasting Task

Imagine: A cafeteria observed sales of 90, 110, and 130 meals. Tomorrow's promotion is already announced.

1. State the target, origin, horizon, and unit.
2. Identify usable information; exclude tomorrow's realized foot traffic.
3. Calculate latest-value and three-observation-mean forecasts.
4. If tomorrow's sales are 120, calculate both errors.

Check

Forecasts: 130 and 110. Errors: -10 and $+10$. Both absolute errors are 10 meals on this day.

From a Forecasting Rule to a Trustworthy Test

- Specify the target, origin, horizon, and available information.
- Arrange observations by entity and date.
- Calculate simple forecasts before fitting more flexible rules.
- Save predictions and compare their errors on the same task.
- Explain uncertainty and the decision a forecast supports.

Key message: Lecture 2 asks when a forecast's apparent success can be misleading.